

3-Month ML / DL / Data Analysis Plan

From scratch to deployed model • ~1.5 hrs/day • 12 weeks

● Theory / concepts

● Hands-on coding

● Project / practice

▲ Milestone

Month 1 — Foundations + Data Analysis

Goal: Python fluency, maths basics, full EDA workflow •

~1.5 hrs/day

Week 1

~10 hrs

Python + NumPy + environment setup

- CS50P lectures 0–3 (functions, loops, exceptions)
- NumPy arrays, broadcasting, indexing exercises
- Set up Jupyter / Google Colab and Git repo
- Mini project: manipulate a CSV with NumPy only

[CS50P \(YouTube\)](#) [numpy.org docs](#)

Week 2

~10 hrs

Pandas + data cleaning

- Kaggle Pandas micro-course (all 6 lessons)
- groupby, merge, pivot, handling nulls & types
- Clean a messy real dataset (Kaggle dataset of choice)

[Kaggle: Pandas course](#) [Pandas docs](#)

Week 3

~10 hrs

Visualization + EDA

- Matplotlib, Seaborn, Plotly basics
- Statistical thinking: distributions, correlations, outliers
- Full EDA notebook on Titanic dataset

[Kaggle: Data Viz course](#) [StatQuest](#) [YouTube](#)

Week 4

~10 hrs

Linear algebra + probability crash course

- 3Blue1Brown: Essence of Linear Algebra (all 15 videos)
- Khan Academy: Probability & distributions basics
- Implement matrix ops and dot products in NumPy

[3Blue1Brown](#) [YouTube](#) [Khan Academy](#)

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Month 1 milestone: EDA notebook published on GitHub with 5+ charts and written insights

Month 2 — Classical Machine Learning

Goal: understand + implement core ML algorithms, win a

Kaggle bronze • ~1.5 hrs/day

Week 5

~10 hrs

Linear & logistic regression from scratch

- CS229 Lecture 1–3 (linear regression, gradient descent)
- Build linear regression with NumPy (no sklearn yet)
- Predict house prices (simple version)

[Stanford CS229](#) [YouTube](#) [Geron book ch. 1–4](#)

Week 6

~10 hrs

Decision trees, forests, SVMs

- CS229 Lecture 4–7, StatQuest tree series
- sklearn: DecisionTree, RandomForest, SVC on Iris
- Classification on Heart Disease dataset (Kaggle)

[StatQuest](#) [YouTube](#) [Geron book ch. 5–7](#)

Week 7

~10 hrs

Model evaluation, tuning, pipelines

- Cross-validation, confusion matrix, ROC-AUC
- GridSearchCV, feature importance, sklearn Pipeline
- Feature engineering: encoding, scaling, imputation

[Coursera ML Specialization](#) [sklearn user guide](#)

Week 8

~10 hrs

XGBoost + Kaggle competition

- Ensemble methods: bagging, boosting, stacking
- XGBoost / LightGBM on tabular data
- Submit to Titanic Kaggle competition, study top kernels

[fast.ai](#) [Practical ML Kaggle competitions](#)

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Month 2 milestone: 2 ML projects on GitHub + first Kaggle competition submission

Week 9 ~12 hrs Neural networks from scratch ● Karpathy: makemore series — backprop by hand ● Build MLP in pure NumPy, then replicate in PyTorch ● CS231n Lecture 1–4 (intro, backprop) Karpathy YouTube CS231n YouTube	Week 10 ~12 hrs CNNs + image classification ● CS231n Lecture 5–9 (CNNs, pooling, architectures) ● Train ResNet on CIFAR-10 using PyTorch ● Transfer learning: fine-tune pretrained model on custom images CS231n YouTube PyTorch tutorials
Week 11 ~12 hrs Sequence models + attention ● CS224n Lecture 1–5 (word vectors, RNNs, LSTMs) ● Karpathy: GPT from scratch (nanoGPT video) ● Implement self-attention block in PyTorch CS224n YouTube Karpathy nanoGPT	Week 12 ~12 hrs Capstone project + what's next ● Build one end-to-end project: image classifier or text classifier ● Deploy model via FastAPI or Streamlit (share a live link) ● Write a short blog post explaining your project ● Plan month 4: pick a specialization track (CV / NLP / RL) Streamlit docs HuggingFace Spaces (free)

MILESTONE	Month 3 milestone: deployed ML/DL app live on the web + blog post published
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Daily schedule (1.5 hrs)	0–30 min Watch lecture / read	30–80 min Code along / implement	80–90 min Review + push to GitHub
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